Assignment 3 MATLAB

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%% Question 1

A= [31, 2, 3, 44];
B= [12, 20, 15, 74];

%(a)

S1=dot(A,B);
fprintf('Solution using dot function %f\n',S1)

%(b)

S2=sum(A.*B);
fprintf('Solution using summing the array products %d\n',S2)

%% Question 2

A = [1, 2, 3; 2, 4, 6; 3, 6, 9];

if det(A)~=0
    inv(A)
else
    fprintf('A is a singular matrix')
end

% det(A) is zero this implies the matrix is singular


%% Question 3

%(a)
A=[1 1 1 1; 2 3 1 5; -1 1 -5 3; 3 1 7 -2];
b=[10;31;-2;18];
X=A\b % or % X=inv(A)*b % or % X=linsolve(A,b)

%b)
A=[3 2 -1; 2 3 1; -1 3 2; 1 -1 -7];
b=[10;31;-2;18];
X=A\b


%% Question 4

% (a)
A=[3 2 5; 2 3 -2; 1 1 1; 2 -4 -7];
b=[22;8;6;-27];
X=A\b

% (b)
```

```
A=[3 2 5; 4 5 -2];
b=[22;8];
X=A\b


%% Question 5

% (a)
A = [12, 4; 3,-5];
B = [2, 12; 0, 0];
D1=A*B
D2=B*A

% (b)
A = [1, 2, 3; 2, 4, 6];
B = [-2, 4; 3, 8; 12, 2];
D1=A*B
D2=B*A


%% Question 6,7

% (a)
A=[-2 1 0; 1 1 0; 5 3 -1];
b=[3;10;10];

fprintf('Solution using left division')
tic
X=A\b
toc

fprintf('Solution using inverse')
tic
X=inv(A)*b
toc

%% Question 6,7

% (b)
A=[3 2 1; 4 -1 3];
b=[4;12];

fprintf('Solution using left division')
tic
X=A\b
toc

fprintf('Solution using inverse')
tic
X=inv(A)*b
toc

%% Question 6,7

% (c)
```

```
A=[3 2 5 1; 1 -3 7 1; 2 2 -3 4; 1 1 1 1];
b=[24;12;17;0];

fprintf('Solution using left division')
tic
X=A\b
toc

fprintf('Solution using inverse')
tic
X=inv(A)*b
toc


%% Question 8

% (a)
function FV=future_value(PV,I,n)
  FV=PV*(1+I)^n;
end
%or
% future_value = @(PV,I,n) PV*(1+I)^n;
% (b)
future_value(1000,0.5,10)



%% Question 9
function Temperature_Conversions()
disp('Temperature Conversion Tables:');
disp('--------------------------------------');

% (a)
disp('Fahrenheit to Kelvin:');
disp('--------------------------------------');
Tf=0:200;
Tk=F_to_K(Tf);
       disp('   Fahrenheit      Kelvin');
         for i = 1:length(Tf)
        fprintf('%10.2f     %10.2f\n', Tf(i), Tk(i));
         end
disp('--------------------------------------');

% (b)
disp('Celsius to Rankine:');
disp('--------------------------------------');
Tc=linspace(0,100,25);
Tr=C_to_R(Tc);
       disp('    Celsius       Rankine');
      for i = 1:25
        fprintf('%10.2f     %10.2f\n', Tc(i), Tr(i));
      end
disp('--------------------------------------');

% (c)
```

```matlab
disp('Celsius to Fahrenheit:');
disp('--------------------------------------');
Tc=linspace(0,100,50);
Tf=C_to_F(Tc);
       disp('    Celsius      Fahrenheit');
         for i = 1:length(Tc)
        fprintf('%10.2f     %10.2f\n', Tc(i), Tf(i));
         end
 disp('--------------------------------------');
end

function FK= F_to_K(F)
       R = (F + 459.67);
   FK = 5/9*R;
end

function CR= C_to_R(C)
      F = (9/5*C +32);
  CR= F+459.67;
end

function CF= C_to_F(C)
         CF = 9/5*C +32;
end
Temperature_Conversions()


%% Question 10

% (a)
clc
function h=height(t)
h= -4.9*(t.^2)+100*t+250;
end

%(b)
t=0:.000001:20;
H=height(t);
T=0;
  for i=2:length(t)-1

      if H(i-1)<H(i)&& H(i+1)<H(i)
          T=t(i);
      end

  end

  if T>0
          fprintf('The rocket starts to fall back to the ground at time %f', T)
      else
          fprintf('Increase the range of t.')
  end

% syms t
%   height= @(t) -4.9*(t.^2)+100*t+250;
```

```
%   v=diff(height,t);
%   time=solve(v==0);
% fprintf('The rocket starts to fall back to the ground at time %f\n', time)

%% Question 11

% (a)

my_function=@(x) -x^5-5*x^3-3*x^2+3+exp(x);
xmin=-100;
xmax=100;

% (b)

fmin=fminbnd(my_function,xmin, xmax);
fprintf('The minimum function value between the range %.2f and %.2f is %f',
xmin,xmax,fmin)



%%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% syms x y z w
% eqn1=x+y+z+w==10;
% eqn2=2*x+3*y+z+5*w==31;
% eqn3=-x+y-5*z+3*w==-2;
% eqn4=3*x+y+7*z-2*w==18;
% sol=solve([eqn1,eqn2,eqn3,eqn4],[x,y,z,w]);
% X=[sol.x;sol.y;sol.z;sol.w];
% display(X)
```